THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE UNIT TEST SET 3

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The sum of first 8 positive odd numbers is: 1

(a) 64 **(b)** 65 **(c)** 63 **(d)** 66

2. In $\triangle ABC$, $AB = 7$, $BC = 8$, $AC = 10$. The triangle is: 1

(a) equilateral **(b)** not right-angled **(c)** right-angled **(d)** isosceles right

3. The distance of $(-7, 2)$ from the y-axis is: 1

(a) 9 **(b)** 8 **(c)** 7 **(d)** 6

4. The 10th term of the AP $7, 4, 1, -2, \ldots$ is: 1

(a) -21 **(b)** -19 **(c)** -18 **(d)** -20

5. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 3 : 5$. Then $AE : AC$ is 3: 1

(a) 9 **(b)** 7 **(c)** 10 **(d)** 8

6. The point $(0, 0)$ is equidistant from $(1, 4)$ and $(5, 2)$ for $x =$ 1

(a) $\frac{3}{2}$ **(b)** $\frac{5}{2}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{7}{2}$

7. The 12th term of $-5, -3, -1, 1, \ldots$ is: 1

(a) 17 **(b)** 19 **(c)** 16 **(d)** 18

8. In $\triangle ABC$, $DE \parallel BC$, $AD = 5$ cm, $DB = 4$ cm, $AE = 15$ cm. Then EC (in cm) $=$ 1

(a) 13 **(b)** 11 **(c)** 14 **(d)** 12

9. The distance between $(0, 0)$ and $(9, 12)$ is: 1

(a) 17 **(b)** 16 **(c)** 15 **(d)** 14

10. The 6th term from the end of the AP with last term 150 and $d = 5$ is: 1

(a) 124 **(b)** 127 **(c)** 126 **(d)** 125

11. $\triangle ABC \sim \triangle DEF$ with areas 9 cm² and 36 cm². If $AB = 3$ cm, then DE (cm) $=$ 1

(a) 8 **(b)** 7 **(c)** 5 **(d)** 6

12. The point which divides the join of $(-3, 0)$ and $(6, 6)$ in the ratio 1:2 internally is: 1

(a) $(0, 2)$ **(b)** $(6, 6)$ **(c)** $(3, 4)$ **(d)** $(\frac{3}{2}, 3)$

13. If $a_n = 3n + 7$, then the common difference is: 1

(a) 5 **(b)** 4 **(c)** 2 **(d)** 3

14. The length of the diagonal of a square of side 6 cm, squared, is: 1

(a) 74 **(b)** 73 **(c)** 71 **(d)** 72

15. The area of the triangle with vertices $(0, 0)$, $(10, 0)$ and $(0, 3)$ is: 1

(a) 15 **(b)** 14 **(c)** 17 **(d)** 16

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. A ladder 25 m long reaches a window 24 m high. Find how far the foot of the ladder is from the wall. 2

17. Find the distance between the points $(-5,\ 7)$ and $(-1,\ 3)$. 2

18. Find the sum of the first 12 multiples of 5. 2

19. Sides of a triangle are 6 cm, 7 cm, 10 cm. State with reason whether it is right-angled. 2

20. Find the coordinates of the point which divides the join of $(-1,\ 7)$ and $(4,\ -3)$ internally in the ratio $2 : 3$. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the area of the triangle with vertices $(2,\ 3)$, $(-1,\ 0)$ and $(2,\ -4)$. 3

22. Find the sum of the first 25 terms of the AP $-9, -7, -5, -3, \ldots$ 3

23. In $\triangle ABC$, $DE \parallel BC$ with $AD : DB = 2 : 3$. Find the ratio of the area of $\triangle ADE$ to that of $\triangle ABC$, and to that of trapezium $DBCE$. 3

24. In what ratio does the x-axis divide the segment joining $(2,\ -3)$ and $(6,\ 9)$? Find the point of division. 3

OR

In an AP, the 4th term is 16 and the 6th term is 28. Find the AP and its 9th term. 3

25. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 2$ cm, $DB = 7$ cm and $AE = 10$ cm, find EC. 3

OR

Find the value of k for which the points $(-1,\ 3)$, $(1,\ 7)$ and $(3,\ k)$ are collinear. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Find the point on the x-axis which is equidistant from $(2,\ 3)$ and $(8,\ 1)$. 5

OR

A loan is repaid in 12 monthly instalments; the first is ₹200 and each later instalment is ₹15 more than the previous. Find the instalment in the last month and the total amount repaid. 5

27. Two similar triangles have perimeters 20 cm and 60 cm. If a side of the smaller is 6 cm, find the corresponding side of the larger and the ratio of their areas. 5

OR

Three vertices of a parallelogram $ABCD$ taken in order are $A(2,\ 1)$, $B(6,\ 2)$ and $C(7,\ 6)$. Find the fourth vertex D. 5

END OF QUESTION PAPER — **Total: 50 Marks**